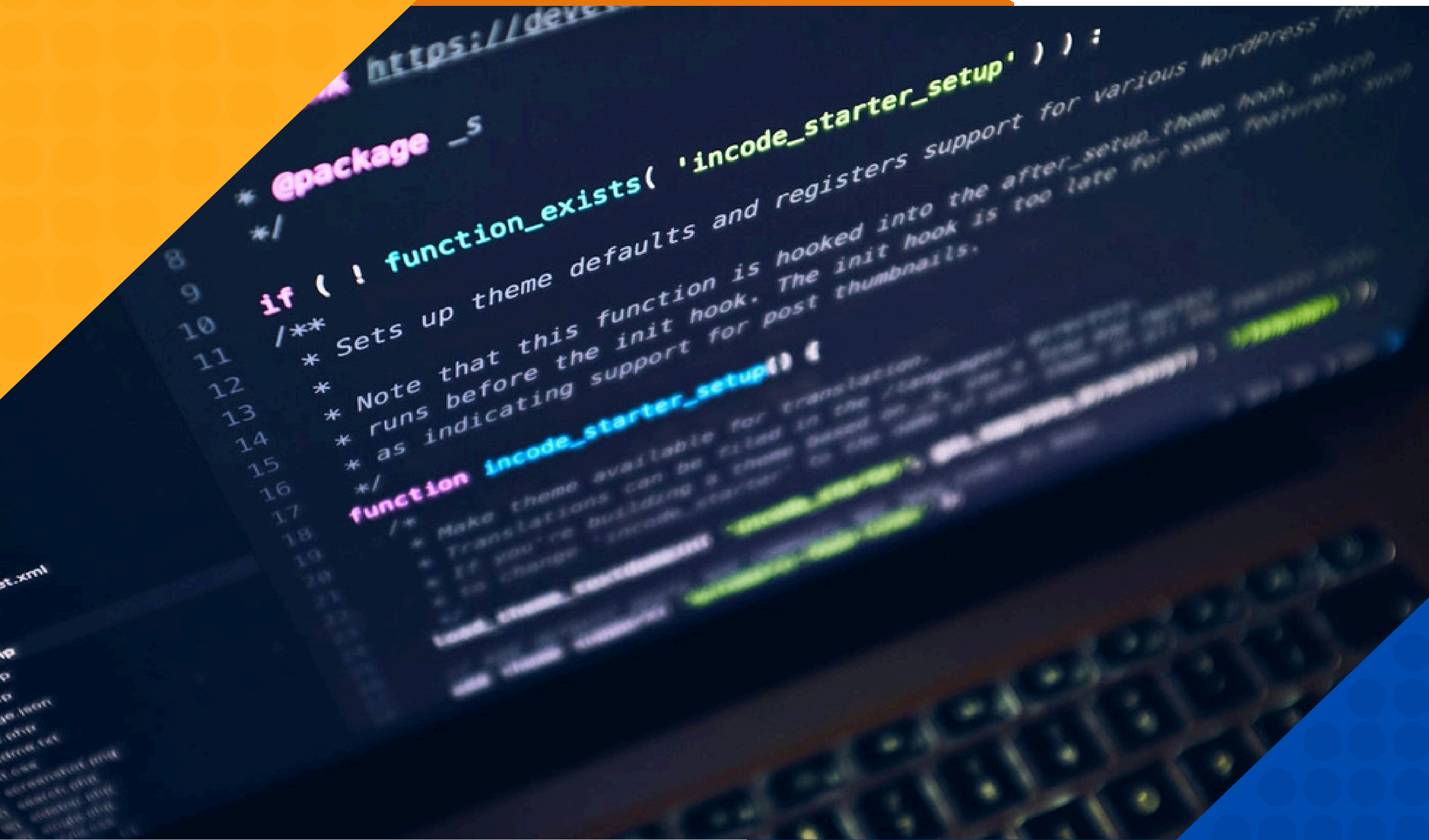

DATA STRUCTURES AND ALGORITHMS SYLLABUS

DURATION - 2 MONTHS
MODE OF TRAINING – OFFLINE/ONLINE

Data Structure & Algorithm

Introduction to Data Structure

- Understanding Data types
- What is Data Structure
- Need of Data Structure
- The Mathematical model

Algorithms

- Understanding algorithms
- How to write algorithms
- Optimizing algorithms
- Finding time complexity of an Algorithm

Lists

- what is List and what is it’s need
- Squentails lists (Arrays) ,Advantages and Limitations
- Implementing sequential lists

Linked Lists

- The LinkedList structure
- Advantages, Limitations
- Singly Linked List
- Doubly Linked List
- Circular Linked List
- Time complexity of Linked List and
- Sequential lists

Stacks

- Understanding stacks
- Stack usage
- Implementing Stack
- Sequential implementation
- Linked implementation
- Double stack and it’s implementatio

Data Structure & Algorithm

Tree

- Introduction to Tree data structure
- Tree usage
- Types of Trees
- General Tree
- Binary Tree
- Binary Search Tree
- Binary Search Tree (BST)
- Understanding Binary Search Tree
- BST usage
- Insertion and deletion
- Understanding BST algorithms
- Inorder, Preorder, Postorder
- BFS and DFS
- Time complexity of BST algorithms
- Implementing BST using arrays
- Constructing BST back using the tree traversals
- Threaded BST
- Why Threaded BST
- Understanding threaded BST
- Threaded BST Algorithms
- Insertion and deletion
- Inorder, Preorder, Postorder
- Time Complexity of Threaded BST
- Height Balanced Trees (AVL Trees)
- What is AVL Tree
- Balance factor, right heavy and left heavy tree
- Height balancing algorithm
- Insertion and deletion algorithms
- Few other types of trees
- Strictly binary tree
- Symmetric tree
- Red Black Tree
- B Tree and B+ Tree

Data Structure & Algorithm

Queues

- What is Queue?
- Understanding Queue usage
- Implementing Queues
- Sequential implementation
- Linked implementation
- Circular Queue
- Usage and implementation
- Types of Queue
- Priority queue
- •Double ended queue

Graphs

- Introduction to Graphs
- Types of Graph
- Directed Graph
- Undirected Graph
- Implementing Graphs
- Sequential implementation
- Linked implementation
- Graph Algorithms
- BFS
- DFS
- Shortest Path Algorithm
- Minimal Spanning Tree
- Creating minimal spanning tree from a graph using
- Kruskal's algorithms
- Prim's algorithm

Hash Tables

- The hashing technique
- Understanding Hash tables
- Time complexity of operations on Hash Table
- Collision resolution algorithms
- Rehashing
- Improving performance using Hash Tables

Data Structure & Algorithm

Infix, Prefix and Postfix Expression

- Infix to prefix conversion and it’s evaluation
- Infix to postfix conversion and it’s evaluation

Searching Algorithms

- Linear Search
- Binary Search
- Indexed sequential search
- Fibonacci Search

Sorting Algorithm

- Bubble Sort
- Selection Sort
- Insertion Sort
- Radix Sort
- Merge Sort
- Quick Sort
- Heap Sort

Finding Time and Space Complexity of Algorithms

- Omega notation
- Big O notatios

Key Facture

- Trainer with 12+ Years of Experience
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